

Incidence of Anterior Cruciate Ligament Reconstruction Among Adolescent Females in the United States, 2002 Through 2014

In 1999, a landmark study¹ reported that women were more likely than men to sustain an anterior cruciate ligament (ACL) injury in sex-comparable collegiate sports. Since then, an extensive body of research has focused on understanding and addressing sex disparities in ACL injuries.^{2,3} This study aimed to investigate time trends in ACL reconstruction among commercially insured individuals in the United States while focusing on sex differences among adolescents with the goal of assessing the effectiveness of the intensive research and prevention efforts of the past 15 years in reducing the sex disparity in ACL injury incidence.

Methods | Data were analyzed from the 2002 through 2014 Truven Health Analytics MarketScan Commercial Claims and Encounters database, which contains health care use information for approximately 148 million privately insured individuals younger than 65 years. Anterior cruciate ligament reconstruction was identified by *Current Procedural Terminology* codes (29888, 27428) and *International Classification of Diseases, Ninth Revision (ICD-9)* procedure codes (81.45), and annual rates were calculated within age (<13, 13-17, 18-30, 31-40, and >40 years) and sex strata. The denominator for each year consisted of the total number of person-months of continuous enrollment in that year for all individuals in the database. This study was determined to be exempt by the University of

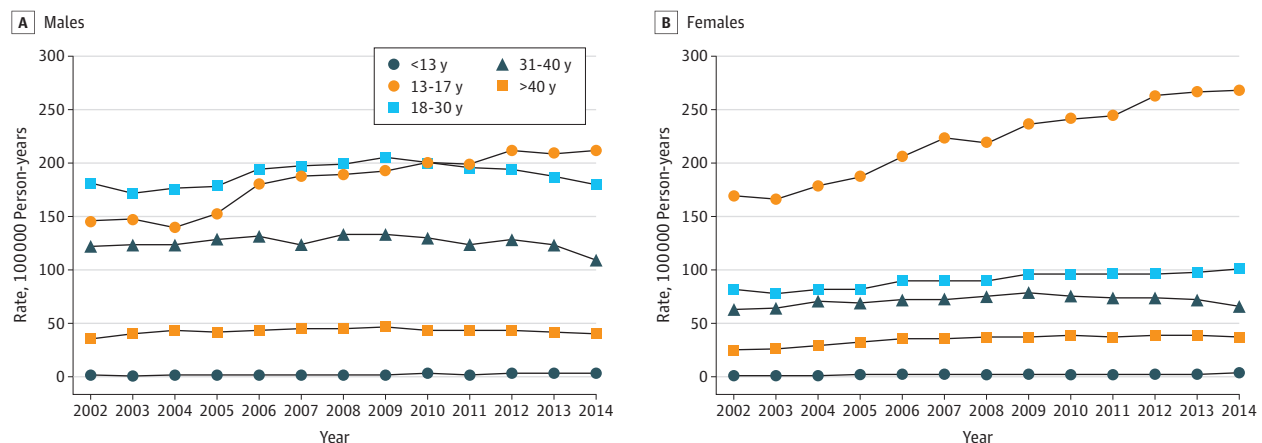
North Carolina at Chapel Hill institutional review board, and patient consent was not required as this study involved existing data.

Our primary analysis estimated incidence rates of ACL reconstruction, a downstream consequence of ACL injuries, because this outcome is more likely to be reliably captured in claims data than an incident ACL injury.⁴ Since trends in ACL reconstruction are not the same as trends in ACL injury incidence, we separately examined whether the proportion of patients with diagnosed ACL injuries who underwent reconstruction within 1 year of receiving their diagnosis changed over time. Statistical comparisons used a χ^2 trend test and Poisson regression with person-month offsets to estimate incidence rate ratios.

Results | There were 283 810 ACL reconstructions and 385 384 623 person-years at risk identified from 2002 through 2014. The rates of ACL reconstruction per 100 000 person-years rose by 22%, from 61.4 in 2002 to 74.6 in 2014 ($P < .01$). Males ($n = 162\,399$) had higher rates of ACL reconstruction than females ($n = 121\,411$) (87.0 vs 61.1 per 100 000 person-years; $P < .01$), but females experienced a more rapid increase in rates than males (34% vs 13%, $P < .01$).

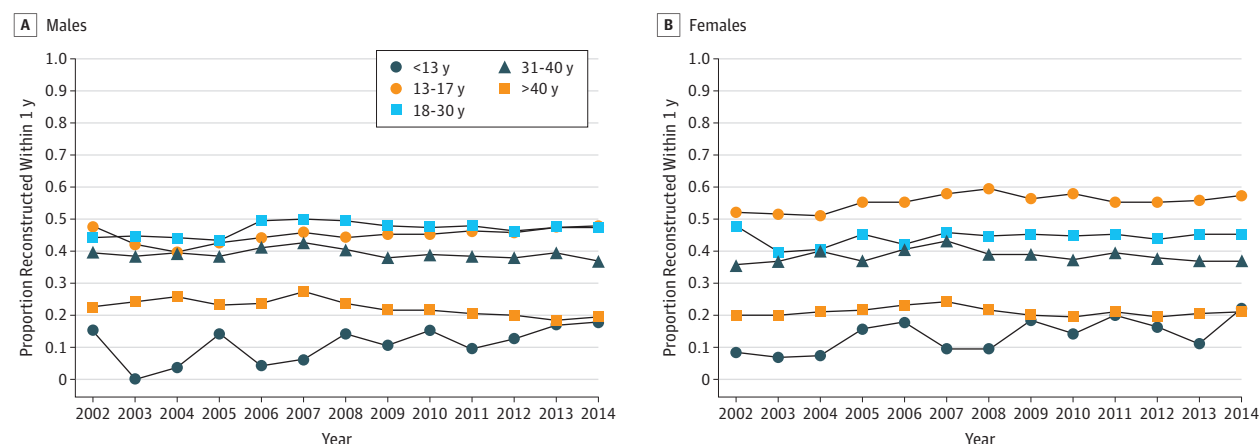
Females and males aged 13 through 17 years had the greatest absolute increases in ACL reconstruction rates over the 13-year study period (females: 169.0 to 268.7; males: 146.8 to 211.7) (**Figure 1**). The rise in rates among female ages 13 through 17 years was so steep that, by the end of the study period in 2014, they had a higher rate of ACL reconstruction than any other age-sex strata. By contrast, at the start of the

Figure 1. Annual Trends in Incidence Rate of Anterior Cruciate Ligament Reconstruction by Age Group



Comparing males ($n = 162\,399$) (A) and females ($n = 121\,411$) (B) included in the Truven Health Analytics MarketScan Commercial Claims database from 2002 to 2014 ($n = 283\,810$).

Figure 2. Annual Trends in Proportion of Individuals With a New Anterior Cruciate Ligament (ACL) Injury Diagnosis Who Underwent ACL Reconstruction Within 1 Year by Age Group



Comparing males ($n = 116\,824$) (A) and females ($n = 86\,060$) (B) included in the Truven Health Analytics MarketScan Commercial Claims database 2002 to 2013 ($n = 202\,884$). New diagnoses of ACL injury were identified by 2 instances: an *International Classification of Diseases, Ninth Revision (ICD-9)* 717.83 or 844.2 diagnosis code within 1 year with no *ICD-9* diagnosis codes within the 180 days

before the first recognized code for ACL injury. One year of continuous enrollment in the database beginning at the first instance of the *ICD-9* diagnosis code was also required to determine if the patient underwent ACL reconstruction within 1 year.

study period in 2002, their rate was comparable with males aged 13 through 17 years and lower than males aged 18 through 30 years.

When limiting analyses to the 202 884 ACL injury diagnoses that had 1 year of continuous enrollment in the database, there was no significant change in the proportion reconstructed over time for females aged 13 through 17 years ($P = .13$) (Figure 2). Changes in surgical practice are therefore an unlikely explanation for the observed increase in 13- through 17-year-old females.

Discussion | Despite intensive research efforts to address ACL injuries among high-risk females, the incidence of ACL reconstruction among 13- through 17-year-old females has continued to increase. Anterior cruciate ligament reconstruction rates for 13- through 17-year-old males are also rising. Multiple clinical trials demonstrate the preventability of ACL injuries.^{2,3} We need to adopt a new paradigm to translate promising evidence from randomized controlled trials^{2,3} into tangible large-scale population-based injury prevention programs. There is a pressing need for research on strategies that stimulate the widespread adoption, implementation, and maintenance of community-focused ACL injury programs.^{5,6} The public health gains from recent increases in athletic participation, particularly by women, will be negated if we fail to address the growing burden of sports-related ACL injuries among adolescents.

Mackenzie M. Herzog, MPH
Stephen W. Marshall, PhD
Jennifer L. Lund, PhD
Virginia Pate, MS
Christina D. Mack, PhD
Jeffrey T. Spang, MD

Author Affiliations: Gillings School of Global Public Health, Department of Epidemiology, University of North Carolina at Chapel Hill, Chapel Hill (Herzog, Marshall, Lund, Pate, Mack); Injury Prevention Research Center, University of North Carolina, Chapel Hill (Herzog, Marshall); QuintilesIMS, Real-World Insights, Research Triangle Park, North Carolina (Herzog, Mack); College of Arts and Sciences, Department of Exercise and Sport Science, University of North Carolina at Chapel Hill, Chapel Hill (Marshall); School of Medicine, Department of Orthopaedics, University of North Carolina at Chapel Hill, Chapel Hill (Spang).

Corresponding Author: Mackenzie M. Herzog, MPH, Department of Epidemiology, University of North Carolina at Chapel Hill, McGavran Greenberg Hall, CB #7435, Chapel Hill, NC 27599-7435 (mherzog@email.unc.edu).

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Concept and design: Herzog, Marshall, Lund, Pate, Spang.

Acquisition, analysis, or interpretation of data: Herzog, Marshall, Lund, Pate, Mack.

Drafting of the manuscript: Herzog, Marshall.

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1. Arendt EA, Agel J, Dick R. Anterior cruciate ligament injury patterns among collegiate men and women. *J Athl Train*. 1999;34(2):86-92.

2. Gagnier JJ, Morgenstern H, Chess L. Interventions designed to prevent anterior cruciate ligament injuries in adolescents and adults: a systematic review and meta-analysis. *Am J Sports Med*. 2013;41(8):1952-1962.

3. Donnell-Fink LA, Klara K, Collins JE, et al. Effectiveness of knee injury and anterior cruciate ligament tear prevention programs: a meta-analysis. *PLoS One*. 2015;10(12):e0144063.
4. Singh JA, Kundukulam JA, Bhandari M. A systematic review of validated methods for identifying orthopedic implant removal and revision using administrative data. *Pharmacoepidemiol Drug Saf*. 2012;21(suppl 1):265-273.
5. O'Brien J, Finch CF. The implementation of musculoskeletal injury-prevention exercise programmes in team ball sports: a systematic review employing the RE-AIM framework. *Sports Med*. 2014;44(9):1305-1318.
6. Finch CF. No longer lost in translation: the art and science of sports injury prevention implementation research. *Br J Sports Med*. 2011;45(16):1253-1257.